What the readings change

A landscape assessment needs relationships as well as area totals.

The main lesson

The papers shift attention from isolated monuments and water polygons towards combinations of landform, water management, productive land and dispersed settlement. A reservoir can store water, receive drainage, obstruct runoff or support several activities. Fields can retain rainfall, use receding floods or receive redistributed water. Similar geometry does not establish the same function.

A disagreement to retain

Van Liere questions strong versions of the hydraulic-city interpretation and stresses local agricultural practice. Groslier reconstructs successive large-scale hydraulic systems. Evans et al. provide later mapping evidence for an extensive integrated network, while Hawken and Cobo Castillo distinguish field patches, gardens and community-managed productive space. These positions overlap in some respects but do not resolve into one agreed model.

How this connects to the present task

Keep the existing approximately 2,000 km² boxes for comparable satellite statistics. Add a separate landscape context that includes relevant catchments, outlets and dispersed features. For each proposed relationship, distinguish what is visible, what terrain makes plausible and what historical or archaeological evidence supports. Modern seasonal water connectivity could contribute to that assessment, but it has not yet been analysed and would not itself establish an ancient connection.

Reading coverage and source limits

These notes cover the main text of the six supplied works, with relevant figures and maps consulted. The Groslier review used local French OCR and selected checks against the scan; it is an analytical summary, not a certified translation. The supplied Van Liere 1989 scan duplicates printed pages 146-147 and omits 144-145: those missing pages have not been reviewed. The Evans lidar file is a manuscript with unverified publication details. Bibliographies guide further reading; inclusion does not mean that every cited work has been read.

Status

This is a completed set of analytical notes for the available supplied material. Portable GIS packages now exist for all four regions, each with exploratory landform layers (hillshade, contours, modelled flow paths and height above drainage) derived from the Copernicus surface model, and an OpenStreetMap temple layer. Native DEM review, catchment-wide extents, agricultural classification, feature dating and independent accuracy assessment remain to be done. Page references below use printed pages, except the Evans manuscript, which uses PDF pages.

Van Liere: water practice and landform

Traditional Water Management in the Lower Mekong Basin | 1980

Argument and evidence

Van Liere distinguishes river channels and levees, backswamps, alluvial plains, terraces and hills, then relates water-management practices to those settings. Floodplain cultivation, bunded fields and flood-recession cultivation use water differently. Bunds can retain, retard, spread or deflect water; their presence is not sufficient evidence of a formal irrigation network (pp. 265-273; Figs. 2, 4-7).

Storage is not the same as irrigation

In his discussion of thamnup/tnub-type works, retained water and receding inundation support a sequence of cultivation opportunities. He argues that some storage was insufficient for downstream irrigation during the dry period. This is an interpretation of particular arrangements, rather than a universal test for every embankment or pond (pp. 272-273). Hawken and Cobo Castillo later explicitly challenge the insufficiency argument.

Where the interpretation is contested

The Angkor discussion strongly separates monumental or religious hydraulics from practical agricultural management (pp. 274-279). Later mapping and agrarian scholarship complicate that division. Neither the religious association of a water feature nor its visibility in imagery establishes that it lacked a productive role. Conversely, a large reservoir is not proof that all nearby cultivation was irrigated.

A useful example

The epilogue describes farmers adapting a modern West Baray scheme towards flood-recession cultivation rather than simply following the intended irrigation layout (p. 279). The point for our assessment is to examine actual seasonal use, not infer practice from the engineering plan alone.

Mapping decision and question

Describe the landform and observed water regime before assigning an agricultural function. Keep rain retention, flood recession and redistributed water as competing explanations where evidence is weak. Ask: which local combinations of field boundaries, elevation and seasonal observations could help distinguish these practices? Sambor Prei Kuk/Isanapura in the paper must not be substituted for Srei Santhor.

Van Liere, W. J. (1980). Traditional Water Management in the Lower Mekong Basin. World Archaeology 11(3), 265-280.

Van Liere: different landscape settings

Mon-Khmer Approaches to the Environment | 1989

What the available chapter argues

Van Liere relates historical settlement and cultivation patterns to environmental settings, contrasting flood-dependent agro-cities with later bunded field blocks, ponds and retained forest. These are broad historical interpretations, not classes that can be identified reliably from a contemporary satellite image alone (available pp. 142-143, 146-159).

Fields can also change hydrology

The chapter treats fields as small water-retaining compartments that alter runoff, sediment movement and channel development, rather than passive land cover (around p. 151). It considers environmental change while rejecting a simple explanation of historical decline through soil poverty. These mechanisms need local evidence before they can explain a specific mapped pattern.

The useful Sukhothai comparison

The discussion and Figs. 9.6-9.7 distinguish low terraces or a fossil alluvial fan from Yom River lowlands. Hill-fed supply, seasonal west-to-east streams, embankments and the watering of seedbeds suggest several scales and purposes of water management (pp. 152-154). This is a stronger starting point than treating the whole Sukhothai region as a flat irrigated plain.

Interpretive limits

Raised linears may be interpreted as roads rather than major canals in this account. Later studies allow multiple functions for such infrastructure. The chapter's historical and ethnic generalisations also require independent support; square fields cannot establish the identity or political organisation of their makers (pp. 154-158). The Isanapura example concerns Sambor Prei Kuk.

Mapping decision and source gap

For Sukhothai, inspect hills, fan/terrace surfaces and river lowlands separately, recording water sources and possible outlets. Do not transfer the Angkor interpretation unchanged. The supplied PDF repeats printed pages 146-147 in consecutive spreads and lacks 144-145. Notes cover all available chapter pages; the missing spread should be obtained before claiming coverage of the complete original chapter.

Van Liere, W. J. (1989). Mon-Khmer Approaches to the Environment. Chapter 9 in Culture and Environment in Thailand, Siam Society symposium volume, pp. 142-159; supplied scan incomplete.

Evans et al.: mapping the wider system

A comprehensive archaeological map of Angkor | 2007

Extent follows the landscape question

The authors combine historical inventories and maps, aerial photography, optical imagery, airborne radar and field/aerial surveys. Their 2,848 km² study area follows watershed boundaries rather than a radius around Angkor Wat. Dispersed temples, occupation mounds, ponds, fields and linear infrastructure form part of the settlement (pp. 14277-14279; Fig. 2).

A concrete evidence rule

Mapping required detection in at least two different data sources, or verification through ground or low-level aerial survey (p. 14279). This offers a useful model for recording evidence, but copies or visual derivatives of the same source are not independent corroboration. Even approximately 5 m AIRSAR data did not consistently resolve all small features. Our 20 m analysis grid and approximately 30 m DEM source cannot inherit their feature-level confidence.

Why seasonal signatures might matter

Radar contrast can reflect rice maturity, soil moisture, relief and the reflective behaviour of bunds over remnant features (p. 14278). This supports looking at temporal and contextual patterns, not equating radar darkness with water in every setting. Sentinel-1 has not yet been used in our assessment; the paper's multi-band airborne radar is not interchangeable with Sentinel-1.

Function and chronology need separate evidence

Roads, canals and embankments can have multiple roles (p. 14280). Figure 4 uses one feature cutting or obscuring another to infer relative chronology. Figure 5 identifies the Krol Romeas distribution outlet on the East Baray's east bank, cautioning against assuming all reservoirs drained through a southwest outlet. Inscriptions, artefacts and architecture provide chronological anchors (p. 14281).

Mapping decision and question

Add a wider catchment context and a feature register with source, observation scale, evidence type and confidence. Retain a neutral linear-feature label when function is unresolved. Ask: can an existing documented connection be recognised in our contemporary imagery, and what remains below its resolution? Their evidence for an integrated hydraulic network does not date every surviving feature or demonstrate its current productive benefit.

Evans, D., et al. (2007). A comprehensive archaeological map of the world's largest preindustrial settlement complex at Angkor, Cambodia. PNAS 104(36), 14277-14282. https://doi.org/10.1073/pnas.0702525104

Groslier: storage, drainage and levels

La cité hydraulique angkorienne: exploitation ou surexploitation du sol? | 1979

What the reconstruction does

Groslier reads the terrain between the Kulen uplands and Tonle Sap through rivers, storage basins, embankments, temples and fields. Water management includes buffering an uneven monsoon, not simply maximising standing water or assuming continuous irrigation (pp. 161-171). Maps 2-6 reconstruct successive configurations: they are not an inventory of features all operating together.

Relative levels are essential; historical heights need care

His source-to-storage-to-downstream reasoning depends on levels and plausible flow paths. He also notes discrepancies between map heights following a change in the Cambodian levelling reference (p. 163, footnote). The reconstructed archaeological lake margin around +16 m must therefore not become a modern EGM2008 DSM threshold. Its date, datum and shoreline interpretation all need checking.

Drainage is as important as supply

Successive systems reuse or obstruct older works; raised roads and banks can redirect runoff (pp. 172-187). Water arriving at the wrong time can be harmful, and excess water must leave fields as well as reach them (pp. 187-189). An inherited structure can support production, constrain it or do both. Observed seasonal water expansion alone does not establish which.

What remains a hypothesis

Yield, cultivation-area and population estimates rely on modern analogues and strong assumptions, including how much land was cultivated (pp. 190-191). Clearing, sedimentation, channel incision and reduced hydraulic head enter his explanation of decline alongside other uncertainties and social factors (pp. 192-195). These are historical arguments, not values or causal conclusions to import into our 2025 results.

Maps to revisit alongside the GIS

Map 1, p. 196: setting; map 2, p. 197: Hariharalaya; maps 3-6, pp. 198-201: Yasodharapura I-IV; map 7, p. 202: regional context and reconstructed lake margin. Compare proposed routes with later mapping, including eastward outlets reported by Evans et al., rather than tracing Groslier's reconstruction as verified drainage. French main text was reviewed with OCR assistance and selected visual checks; numerical or translated quotations require renewed checking against the scan.

Groslier, B.-P. (1979). La cité hydraulique angkorienne: exploitation ou surexploitation du sol? Bulletin de l'École française d'Extrême-Orient 66, 161-202.

Hawken and Cobo Castillo: the mosaic

Angkor's Agrarian Economy: A Socio-Ecological Mosaic | Chapter 19

Productive space is part of the city

Rice fields, gardens, orchards and managed forests are components of dispersed urban settlement. The chapter emphasises diversity and climatic risk buffering, rather than treating agrarian land as an empty hinterland or assuming maximum yield was the only aim (pp. 338-344). A present-day cropland mask cannot represent all of these productive spaces.

Water regimes and a real disagreement

Flood-recession, floating/deep-water and rainfed lowland rice, together with upland gardens, occupy different settings (pp. 340-341). The authors explicitly dispute Van Liere's claim that thamnup storage was insufficient for downstream irrigation. The practical lesson is to test a local water arrangement, rather than turning either position into a universal rule.

Patches offer a better unit for comparison

A patch is identified through combinations of morphology, orientation, gauge, proximity and association. At Pre Rup, smaller patches form a matrix with differing alignments and temple associations; embankments complicate southward water movement. At Phnom Bok, the authors describe a more coherent ordered field system, with small channels distributing water westward and excess water moving south (pp. 344-350; Figs. 19.2-19.3). These are documented case interpretations, not areas newly measured in this project.

Gardens need different evidence

The later vignettes bring mapping, archaeological plant remains and ethnographic knowledge together around gardens and arboreal landscapes (pp. 351-355). Table 19.1 records recovered plant taxa, not crop abundance or yield; a recovered plant need not have grown at the exact excavation point. The conclusion notes insufficient archaeobotanical evidence to reconstruct field water regimes, with weed flora a promising source of evidence (p. 355).

Mapping decision and question

Begin with one documented field patch and its landform, water sources, outlets and settlement associations. Compare modern seasonal observations within that unit. Keep FTW modelled field outlines separate from historically interpreted patches. Ask whether present water or vegetation differences align with the published patch structure; do not label the patches ancient, irrigated or still productive from alignment alone.

Hawken, S., and Cobo Castillo, C. Angkor's Agrarian Economy: A Socio-Ecological Mosaic. Supplied chapter 19, pp. 338-359; main text pp. 338-355. https://doi.org/10.4324/9781351128940-24

Evans: evidence, scale and interpretation

The Predicament of Archaeological Lidar: Perspectives from Southeast Asia | Supplied manuscript

Match the data to the question

Lidar reveals some material traces exceptionally well, but coverage, feature size, vegetation, processing and prior knowledge shape what becomes visible. Acquisition density and ground filtering need to suit the actual features; aggressive filtering can remove cultural detail. Aerial and ground observations both have biases (PDF pp. 17-28). Our Copernicus surface model is not a substitute for archaeological bare-earth lidar.

Validation is not simply visiting every feature

The manuscript challenges treating new ground observation as the only legitimate reference. Existing surveys and documented archaeological knowledge can be appropriate evidence for a specific question. Targeted follow-up should be chosen for the additional information it can provide (PDF pp. 17-21, 27-28). This does not validate our present classifications: the suitability, age and scale of any reference still need assessment.

Images can contain chronological relationships

Superposition, alignments and spatial relationships can support relative chronology when combined with dated anchors (PDF pp. 21-23). It would be too strong to say imagery can never inform dating. However, our present optical composites, unanchored modelled field outlines and coarse heights are not an ancient/modern dating system.

Make interpretation reproducible

Record the interpreter, source data, visualisation, working scale and supporting evidence, with confidence appropriate to the feature class and visibility (PDF p. 26). Automation must be assessed for transfer between landscapes and for the quality of training and reference data, rather than assumed to generalise from a recognisable shape (PDF pp. 28-32).

Landscape inheritance is not automatically a benefit

Durable material arrangements can outlast their original meaning and constrain later activity as well as support it. Broader social and ecological explanations require more than mapped traces (PDF pp. 32-38). The manuscript also discusses local participation, access and research infrastructure (PDF pp. 1-16). For this task, separate a surviving feature, a current physical relationship and evidence of continued productive service. Publication details remain unverified; the filename's year is not treated as a publication date.

Evans, D. (n.d.). The Predicament of Archaeological Lidar: Perspectives from Southeast Asia. Supplied 56-page manuscript; main text and acknowledgements to PDF p. 38, followed by references.

From reading to a testable assessment

A practical sequence for completing the professor's landscape task

1. Choose a defensible landscape unit

Use the published Pre Rup or Phnom Bok examples as candidates for discussion, not as an already selected or digitised study boundary. Check access to the underlying mapping and its terms. Retain the earlier comparison box, but add the headwaters, storage features and outlets relevant to the chosen unit. Repeat site-specific reasoning for the other regions; Sukhothai and Java require different terrain contexts.

2. Build an evidence register

For each feature record geometry, name/identifier, source and page/map, observation period, working scale, proposed function, chronological evidence, confidence and unresolved alternatives. Separate observed form from interpreted function and date. Historic maps, modern map labels and remotely sensed features should remain distinguishable. Do not infer antiquity from wat/candi names or present field geometry.

3. Check terrain before drawing flow arrows

Retrieve native elevation data and quality masks over the wider context. Review datum, units, canopy/building effects, water editing, tile seams and zero/nodata handling. Compare an additional suitable elevation source without assuming independence or averaging by default. Preserve original and conditioned surfaces, then test broad profiles, outlets and sensitivity to conditioning.

4. Test present seasonal relationships

First ask whether known field and hydraulic arrangements correspond to contemporary water and vegetation patterns. Distinguish simple proximity, contiguous surface-water patches, terrain-plausible routes and historically documented connections. Image adjacency is not demonstrated flow: culverts, narrow canals, embankments and under-canopy water may be unresolved. Sentinel-1 time-series connectivity is an option to evaluate, not a result already obtained.

5. Report what the evidence supports

Retain observation coverage and uncertainty, sample against suitable independent references, and show examples where interpretation fails. Landesque capital requires evidence of inherited material arrangements and present productive service; neither is established by seasonal greening alone. A small documented case is a useful first test. It complements the requested regional GIS packages and does not replace their completion.

Terrain audit and remaining work

Measurements from the existing cached rasters | Not a new elevation survey

What was measured

The table summarises the original analytical boxes. P05 and P95 are elevation percentiles; their difference describes broad variation, not terrain accuracy. Values come from Copernicus GLO-30 surface heights resampled to a 20 m grid. The 20 m spacing does not add native terrain detail.

Region	P05 (m)	Median (m)	P95 (m)	P95-P05 (m)
Angkor	5.3	17.0	55.6	50.3
Srei Santhor	3.6	8.3	13.2	9.6
Sukhothai	45.1	60.0	380.3	335.2
Prambanan	52.1	191.4	748.2	696.1

What needs checking

Prambanan has the greatest broad variation in these boxes, but Sukhothai also contains substantial high ground. Srei Santhor's negative tail reaches about -17 m; investigate water editing, masks and an additional terrain source. Negative values alone do not establish an error. Finite coverage throughout each box does not establish accurate heights.

Before routing or fusion

Product documentation gives EGM2008 heights in metres; local rasters encode horizontal UTM CRSs only. No vertical transformation or local elevation validation has been performed. Cached tile files are already warped, and earlier preparation discarded zero-valued samples. Retrieve native data and masks before using the surface to model drainage or fusing it with another source.

Progress and remaining work

Completed: the four-region exploratory assessment, and portable GIS packages for Angkor, Srei Santhor, Sukhothai and Prambanan. Each package includes hillshade, smoothed-DSM contours, modelled flow paths (>=5 km2), height above modelled drainage, and OpenStreetMap temple, wat/candi, ruin and heritage points (434, 187, 483 and 571 points respectively). OpenStreetMap coverage is uneven: it records no archaeological sites at all in the Srei Santhor window, so absence of points is not absence of sites. The terrain layers are exploratory surface-model derivatives, not a validated drainage model. Still to do: wider landscape extents that include contributing catchments, comparison with a second elevation source, a sourced archaeological-feature register beyond OpenStreetMap, and independent accuracy assessment. Accepted 2025 agricultural, rice and irrigated areas remain unestimated.

Terrain source specifications: https://dataspace.copernicus.eu/explore-data/data-collections/copernicus-contributing-missions/collections-description/COP-DEM. Product-level accuracy specifications are not local error measurements.